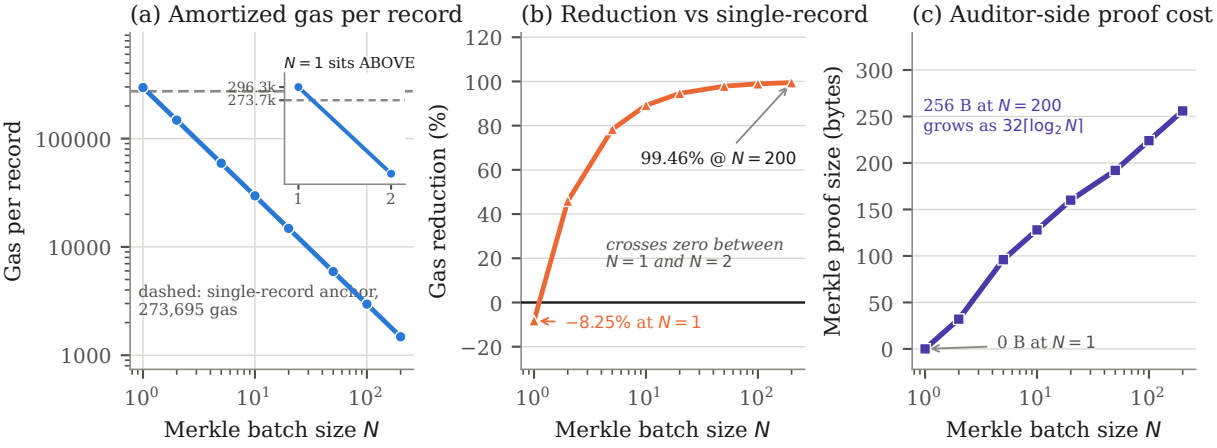
(a) Amortized gas per record
Gas per record
100000
10000
1000
10^0
10^1
10^2
Merkle batch size N
$N = 1$ sits ABOVE
296.3k
273.7k
1
2
dashed: single-record anchor,
273,695 gas
(b) Reduction vs single-record
Gas reduction (%)
120
100
80
60
40
20
0
−20
99.46% @ $N = 200$
crosses zero between
$N = 1$ and $N = 2$
−8.25% at $N = 1$
Merkle batch size N
(c) Auditor-side proof cost
Merkle proof size (bytes)
300
250
200
150
100
50
0
256 B at $N = 200$
grows as $32\lceil \log_2 N \rceil$
0 B at $N = 1$
Merkle batch size N